

Notes on Variance Calculations

The mathematical derivations showing how we can set achieve the following in our simulations:

- The variance of X , Y and C is 1.
- Each variable has a mean 0.
- The autoregressive effects of X_{t-1} on X_t is 0.25 (same for Y_{t-1} on Y_t).
- The lag-1 cross effect of X_{t-1} on Y_t is 0.10 (same for Y_{t-1} on X_t).
- The confounder C has an effect of 0.15 on X_t and an effect of 0.20 on Y_t .
- The innovations have residual correlations of 0.3 at each time-point, but not across time-points.

Defining matrices

We have variables X over occasions T and Y over occasions T . This yields the vector:

$$Z_t = \begin{bmatrix} X_t \\ Y_t \end{bmatrix}$$

For one individual i , the vector looks like:

$$Z_i = \begin{bmatrix} X_{i1} \\ Y_{i1} \\ X_{i2} \\ Y_{i2} \\ X_{i3} \\ Y_{i3} \\ X_{i4} \\ Y_{i4} \\ X_{i5} \\ Y_{i5} \end{bmatrix}.$$

Expanding for N individuals:

$$Z = \begin{bmatrix} X_{1,1} & Y_{1,1} & X_{1,2} & Y_{1,2} & X_{1,3} & Y_{1,3} & X_{1,4} & Y_{1,4} & X_{1,5} & Y_{1,5} \\ X_{2,1} & Y_{2,1} & X_{2,2} & Y_{2,2} & X_{2,3} & Y_{2,3} & X_{2,4} & Y_{2,4} & X_{2,5} & Y_{2,5} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ X_{N,1} & Y_{N,1} & X_{N,2} & Y_{N,2} & X_{N,3} & Y_{N,3} & X_{N,4} & Y_{N,4} & X_{N,5} & Y_{N,5} \end{bmatrix}.$$

Now for time-point t , X_t and Y_t are made up of some part of the confounder effect, some part innovations (unexplained variances), and if $t > 0$, some part of

the dynamic process, i.e. the lagged and autoregressive effects. We can capture this in the formula:

$$Z_t = AZ_{t-1} + BU + E_t,$$

where

- A is the matrix of autoregressive and cross-lagged effects,
- B is the matrix of confounder effects,
- U is the vector of time-invariant confounders,
- E_t is the vector of innovations at time-point t .

If we have 3 confounders, and we choose our effects as above, we get:

$$A = \begin{bmatrix} a_x & b_y \\ b_x & a_y \end{bmatrix}.$$